

Defense Questions and Answer Strategies

答辩可能问题与回答思路

High-Probability Questions 高概率问题

1. Why use MADRL instead of centralized optimization or MPC?

为什么使用 MADRL，而不是直接使用集中式优化或 MPC？

中文回答思路：MPC 确实更适合在给定模型和预测条件下求短期滚动优化，因此本研究并不是为了证明 MADRL 在所有指标上优于 MPC。MADRL 的意义在于训练完成后可以快速执行，并且每个 prosumer 可以根据本地 observation 进行 decentralized decision。它更适合研究 EV charging、battery arbitrage、PV utilization 和 grid feedback 之间的耦合关系。

English answer: MPC is indeed strong for short-term rolling optimization when the model and forecasts are available. The goal of this thesis is not to prove that MADRL outperforms MPC on every metric. Instead, MADRL is used because after training it can execute decisions quickly and in a decentralized way. Each prosumer can act based on its local observation, while the learned policy can still capture the coupling among EV charging, battery arbitrage, PV utilization, and grid feedback.

2. Is your MADRL framework centralized training with decentralized execution?

你的 MADRL 是 centralized training decentralized execution 吗？

中文回答思路：是的。训练阶段 centralized critic 可以利用 joint state 和 joint action 来估计 value，从而学习不同 prosumer 之间通过 power flow 产生的耦合影响。执行阶段每个 actor 只基于本地 observation 和 forecast sequence 输出 action，因此满足 decentralized execution。

English answer: Yes. During training, the centralized critic can use the joint state and joint actions to estimate the state-action value, so it can learn the coupling among prosumers through the power-flow calculation. During execution, each actor only uses its local observation and forecast sequence to output actions. Therefore, the framework follows centralized training and decentralized execution.

3. Why does the soft EV constraint perform poorly? Is it only because the penalty weight is insufficient?

为什么 soft EV constraint 表现不好？只是因为 penalty 权重不够吗？

中文回答思路：不只是权重问题，而是 sparse reward 和 credit assignment 问题。Soft mode 中 departure penalty 主要在离家时刻触发，夜间很多充电决策只有到最后才体现结果。相比之下，battery arbitrage reward 每个 step 都有价格反馈，更容易被 critic 学到。

English answer: It is not only a weight issue. The main problem is sparse reward and credit assignment. In the soft mode, the departure penalty is mainly triggered near the departure time, so many charging decisions

made during the night only receive feedback at the end. In contrast, battery arbitrage provides dense price-related feedback at every time step, so the critic learns it more easily.

4. Why is the Progress penalty more effective than simply increasing the EV cost weight?

为什么 **Progress penalty** 比单纯增加 **EV cost weight** 更有效?

中文回答思路: Cost weight 改变的是 reward scale, 而 Progress penalty 改变的是 reward 的 temporal structure。Progress penalty 在 EV connection window 内构造从 arrival SoC 到 target SoC 的参考轨迹, 把原本只在 departure step 出现的 sparse signal 转换成 dense temporal guidance。

English answer: The cost weight mainly changes the reward scale, while the Progress penalty changes the temporal structure of the reward. It builds a reference trajectory from the arrival SoC to the target SoC within the EV connection window. Therefore, the original sparse departure signal is transformed into dense temporal guidance, making it easier for the critic and actor to learn gradual charging behavior.

5. Why can the hard EV constraint improve departure SoC? Does it damage RL learning?

Hard EV constraint 为什么能改善 departure SoC? 它是否破坏 RL 学习?

中文回答思路: Hard EV constraint 把 EV feasibility 从 reward learning 转移到 action execution layer。每个 step 计算满足 departure target 所需的 minimum charging power, 如果 actor 请求值太低, 环境会修正到最低可行功率。但它也引入 raw action 和 executed action 的 mismatch, 可能影响 actor-critic learning。

English answer: The hard EV constraint moves EV feasibility from reward learning to the action execution layer. At each step, the environment calculates the minimum charging power required to satisfy the departure target. If the actor requests a lower power, the environment corrects it to the minimum feasible value. This improves feasibility, but it also creates a mismatch between the raw actor action and the executed action, which can affect actor-critic learning.

6. What is the essential difference between safety projection and reward penalty?

Safety projection 和 **reward penalty** 的本质区别是什么?

中文回答思路: Reward penalty 是事后惩罚, 允许 unsafe action 执行后通过 reward 扣分学习。Safety projection 是执行前修正, 会直接修改 battery power 或 PV curtailment, 使动作更接近 grid-safe feasible region。

English answer: The reward penalty is a soft constraint. It allows unsafe actions to be executed and penalizes them afterward through the reward. Safety projection is a hard correction mechanism. It modifies the action before execution, for example by adjusting battery power or PV curtailment, so that the final action is closer to the grid-safe feasible region.

Medium-Probability Questions 中概率问题

7. Why does economic performance decrease after projection improves safety?

为什么 **projection** 改善安全性后，经济性会下降？

中文回答思路：Battery arbitrage 依赖精确的充放电时机和功率幅度。Projection 会在 grid 接近安全边界时修正 battery power 或 PV curtailment，因此 actor 原本根据价格信号想执行的动作可能无法完全执行，storage profit 会下降。

English answer: Battery arbitrage depends on precise charging and discharging timing and power magnitude. When the grid approaches safety limits, projection modifies battery power or PV curtailment. Therefore, the action intended by the actor based on the price signal may not be fully executed, reducing storage profit and increasing total cost.

8. What does a flat or non-smooth Q landscape mean for the actor?

Q landscape 出现 **flat region** 或 **non-smooth** 对 **actor** 有什么影响？

中文回答思路：Actor 的更新依赖 critic 对 action 的 value gradient。如果多个 raw action 被 correction/projection 映射成相似 executed action，Q 对 raw action 的变化会变小，形成 flat region。如果 action 被突然 clipping，Q 也可能不平滑。这会让 actor gradient 变弱或不稳定。

English answer: The actor update depends on the critic-estimated value gradient with respect to the action. If many raw actions are mapped to similar executed actions by correction or projection, the Q value changes only weakly with the raw action, creating flat regions. If actions are abruptly clipped or projected, the Q landscape can also become non-smooth. This makes the actor gradient weaker or less stable.

9. Why is the effect of the price-aware penalty limited?

为什么 **price-aware penalty** 的效果有限？

中文回答思路：Price-aware penalty 只改变 reward preference，但 EV charging 的可调度空间已经被 connection window 和 hard feasibility 压缩。如果为了满足 departure SoC 必须充电，即使电价较高也不能完全避免。同时 grid projection 限制 battery 补偿能力，因此边际效果有限。

English answer: The price-aware penalty changes the reward preference, but the schedulable space of EV charging has already been compressed by the connection window and the hard feasibility constraint. If charging is required to satisfy the departure SoC, the agent cannot completely avoid charging even during high-price periods. At the same time, grid projection limits the battery's ability to compensate for the load change. Therefore, the marginal effect is limited.

10. Why choose the 70% quantile as the price-aware threshold?

为什么选择 **70% quantile** 作为 **price-aware threshold**？

中文回答思路：Quantile threshold 比固定价格阈值更鲁棒，因为不同 episode 的价格水平不同。70% 是折中选择：阈值太低会过度惩罚 EV charging，阈值太高则触发太少、学习信号不足。

English answer: A quantile threshold is more robust than a fixed price threshold because the price level can vary across episodes. The 70% quantile is a compromise. If the threshold is too low, too many time steps are penalized and EV charging becomes overly suppressed. If it is too high, the penalty is triggered too rarely and the learning signal becomes weak.

11. Why is V2G not considered? Does this affect the conclusion?

为什么 EV 不考虑 V2G? 这会不会影响结论?

中文回答思路: 本研究把 EV 建模为 flexible load, 而不是 mobile storage, 目的是聚焦 EV charging demand、departure SoC requirement 和 grid safety。V2G 会引入 battery degradation、用户接受度和额外 reward/constraint 设计, 因此属于未来扩展方向。

English answer: This thesis models the EV as a flexible load rather than mobile storage. The goal is to focus on EV charging demand, departure SoC requirements, and grid safety. V2G would introduce additional issues such as battery degradation, user acceptance, and extra reward and constraint design. Therefore, it is a future extension rather than the focus of this thesis.

Critical Questions 尖锐问题

12. Your reward has many penalty terms. Are the results mainly driven by parameter tuning?

你的 reward 有很多 penalty, 结果是不是调参驱动的?

中文回答思路: Reward weight 确实会影响结果, 所以 thesis 不是只展示一个最终 controller, 而是按 Base、Cost Weight、Progress、Projection、Hard Correction、Price-aware 的逻辑逐步加入机制, 分析每个机制的边际影响。

English answer: Reward weights do affect the results, which is common in constrained reinforcement learning. Therefore, the thesis does not only report one final controller. Instead, it follows a development logic: Base, Cost Weight, Progress, Projection, Hard Correction, and Price-aware penalty. This allows the marginal effect of each mechanism on EV feasibility, battery profit, and grid safety to be analyzed.

13. Why not use constrained RL, such as Lagrangian methods or CPO?

为什么不用 constrained RL, 例如 Lagrangian method 或 CPO?

中文回答思路: Constrained RL 是合适的未来方向。当前 thesis 使用 penalty 和 projection, 是因为 grid safety 需要 pandapower power flow 检查, projection 可以在 execution layer 直接修正 unsafe action, 工程上更稳定。

English answer: Constrained RL is a very suitable future direction. In this thesis, penalty and projection are used because grid safety must be evaluated through pandapower power-flow calculations, and projection can directly correct unsafe actions at the execution layer. Lagrangian constrained RL could automatically adjust constraint weights, but it may also introduce extra instability in multi-agent continuous control.

14. Does projection create inconsistency between training and execution?

Projection 会不会导致训练和执行不一致?

中文回答思路: Projection 在训练和测试中都启用, 所以不是 train-test mismatch。但它确实造成 raw action 和 executed action mismatch。Reward 和 next state 由 projected action 决定, 而 actor gradient 仍然作用在 raw action 上, 因此 learning efficiency 会受到影响。

English answer: Projection is enabled during both training and testing, so it is not a train-test mismatch. However, it does create a mismatch between the raw action and the executed action. The reward and next state

are determined by the projected action, while the actor gradient still updates the raw action output. This can reduce learning efficiency and motivates projection-aware training in future work.

15. Are the baselines fair? Do MADRL and MPC use the same forecast information?

Baseline 是否公平? **MADRL** 和 **MPC** 是否使用相同预测信息?

中文回答思路: Baseline 的作用不同。MPC 表示 optimization-based controller 在给定模型和预测下的性能水平, rule-based 表示简单 heuristic。MADRL 和 MPC 都使用未来信息, 但决策方式不同: MPC 显式求解优化问题, MADRL 通过训练后的 policy inference 输出动作。

English answer: The baselines serve different purposes. MPC represents the performance of an optimization-based controller under a given model and forecast, while the rule-based controller represents a simple heuristic. MADRL and MPC both use future information, but their decision mechanisms are different. MPC explicitly solves an optimization problem, while MADRL outputs actions through a trained policy.

16. If MADRL does not clearly outperform MPC, what is your contribution?

如果 **MADRL** 没有明显超过 **MPC**, 你的贡献是什么?

中文回答思路: 贡献不是单纯性能超过 MPC, 而是构建集成 EV、battery、PV、LSTM forecast、pandapower power flow 和 MADRL 的 multi-prosumer framework, 并系统分析不同 constraint-handling mechanisms 对学习行为的影响。

English answer: The contribution is not simply to outperform MPC. The contribution is to build a multi-prosumer control framework integrating EVs, stationary batteries, PV, LSTM forecasts, pandapower power flow, and MADRL training. More importantly, the thesis systematically analyzes how different constraint-handling mechanisms affect learning behavior, including sparse reward, progress guidance, hard correction, and safety projection.

17. Why only three prosumers? Can the method scale?

为什么只有 3 个 **prosumer**? 能否扩展?

中文回答思路: 3 个 **prosumer** 是为了保持实验可解释性, 同时仍能体现 multi-agent coupling。扩展到更多 **prosumer** 可行, 但会增加 joint action/state 维度、non-stationarity 和 grid coupling 难度。未来可使用 parameter sharing、GNN critic 或 mean-field MARL。

English answer: Three prosumers are used to keep the experiments interpretable while still capturing multi-agent coupling. Scaling to more prosumers is possible, but it increases the dimensions of the joint state and action space, strengthens non-stationarity, and makes grid coupling more complex. Future work can use parameter sharing, graph neural network critics, or mean-field MARL to improve scalability.

18. If the actor only uses local observations, how can it handle global grid safety?

Actor 只用 **local observation**, 如何处理 **global grid safety**?

中文回答思路: Local actor 本身不能完全知道全局 power-flow 状态。因此训练时 centralized critic 学习多 agent action 对全局 reward 的影响; 执行时 safety projection 使用 grid-level information 进行修正。经济调度由 actor 学习, 安全性由 reward signal 和 projection 共同处理。

English answer: The local actor cannot fully observe the global power-flow state. Therefore, during training, the centralized critic learns how multi-agent actions affect the global reward. During execution, safety projection uses grid-level information to correct unsafe actions. In this sense, economic scheduling is mainly learned by the actor, while grid safety is handled by both reward feedback and projection.

19. How do LSTM forecast errors affect MADRL performance?

LSTM 预测误差对 MADRL 有什么影响?

中文回答思路: 预测误差会影响 agent 对未来 price、PV、load 的判断。电价峰谷被 smoothing 后, battery arbitrage 和 price-aware EV charging 都会受到影响。未来可以考虑 probabilistic forecasting 或 robust RL。

English answer: Forecast errors affect the agent's understanding of future prices, PV generation, and load demand. If price peaks and valleys are smoothed, battery arbitrage and price-aware EV charging can both be weakened. For example, if low-price periods are overestimated, the agent may miss charging opportunities. Future work could consider probabilistic forecasting or robust RL.

20. Which controller would you recommend in practice?

如果实际应用, 你最推荐哪个 controller?

中文回答思路: 如果经济性优先, 推荐 Base_hard, 因为它保证 EV feasibility 且保留较强 battery arbitrage, 但 grid safety 风险更高。如果安全性和稳健性优先, 推荐 Projection_hard。如果限定 soft framework, 推荐 Base_Progress_soft。

English answer: If economic performance is the priority, I would recommend Base_hard, because it guarantees EV feasibility while preserving stronger battery arbitrage, although grid safety risk remains higher. If safety and robustness are the priority, I would recommend Projection_hard. Within the soft-control framework, I would recommend Base_Progress_soft.

General Answering Framework 万能回答框架

中文: 遇到算法老师追问时, 可以按四步回答:

1. 先承认 trade-off: 这个问题确实存在经济性、安全性和学习稳定性的权衡。
2. 定位机制层级: 这个机制作用在 reward layer、action execution layer, 还是 critic learning layer。
3. 解释算法影响: 它改变的是 reward density、action-reward consistency、actor gradient, 还是 credit assignment。
4. 回到实验结果: 这也是为什么实验中会出现 safety improved, but economic performance decreased。

English: When an algorithm-oriented professor asks a follow-up question, the answer can follow four steps:

1. First acknowledge the trade-off among economy, safety, and learning stability.
2. Identify the mechanism layer: reward layer, action execution layer, or critic learning layer.

3. Explain the algorithmic impact: reward density, action-reward consistency, actor gradient, or credit assignment.
4. Link the explanation back to the experiment, for example: safety improves, but economic performance decreases.

Core thesis line: Soft-control experiments mainly address sparse reward and credit assignment, while hard-control experiments mainly analyze how action correction and projection affect actor-critic learning and economic scheduling.

核心主线： Soft 线主要解决 sparse reward 和 credit assignment；hard 线主要分析 action correction 和 projection 对 actor-critic learning 以及 economic scheduling 的影响。